ADVANCED TEXT GENERATION & SUMMARIZATION

- MODULE 06: CONTROLLING PROBABILITY & EFFICIENT FINE-TUNING (PEFT/LORA)

- THEME: "TAMING THE STOCHASTIC PARROT: CONTROL & EFFICIENCY IN GENERATION"

THE AUTOREGRESSIVE ENGINE

Recall: Language Models predict the *next token* based on probability.

The Formula: P(w_t | w_{1:t-1})

The Dilemma: The model assigns a probability to *every word in the dictionary*.

The Question: How do we choose? Do we always pick the most likely word? (Hint: No).

STRATEGY 1: GREEDY SEARCH & BEAM SEARCH

Greedy Search: Always pick the #1 most likely word.

- Pros: Fast.
- Cons: Repetitive, boring, gets stuck in loops.

Beam Search: Keep the top N sequences alive at each step, choose best total path at end.

- Visual: A tree diagram expanding and pruning branches.
- Use Case: Translation & Summarization (accuracy > creativity).

STRATEGY 2: SAMPLING (THE "CHAOS" KNOBS)

Random Sampling: Pick the next word based on probability distribution.

Temperature (T): Reshapes the probability curve.

- **Low T (< 1.0):** Exaggerates peaks. Conservative & confident. (Coding/Math).
- **High T (> 1.0):** Flattens peaks. Allows rare words. (Poetry/Brainstorming).

STRATEGY 3: TOP-K & NUCLEUS (TOP-P) SAMPLING

- **THE MODERN STANDARD.**
- **TOP-K:** ONLY SAMPLE FROM THE TOP K (E.G., 50) WORDS. CUTS OFF THE "LONG TAIL".
- **NUCLEUS (TOP-P):** DYNAMIC. SAMPLE FROM SMALLEST SET OF WORDS WHERE CUMULATIVE PROB = P (E.G., 0.9).
    - WHY IT WINS: IT ADAPTS. SET IS SMALL IF MODEL IS SURE, LARGE IF UNSURE.

ROBOTICS USE CASE: GENERATING MISSION REPORTS

- **SCENARIO:** A ROBOT EXPLORES A CAVE AND SENDS A TEXT UPDATE.
- **CONSTRAINT:** BANDWIDTH IS LOW. WE NEED CONCISE SUMMARIES.
- **STRATEGY:**
  - RAW LOG: [TIME: 10:00, LOC: X,Y, BAT: 80%, OBJ: ROCK...]
  - GENERATED SUMMARY: "EXPLORED SECTOR 4. FOUND GEOLOGICAL OBSTACLES. BATTERY STABLE."

EXTRACTIVE VS. ABSTRACTIVE SUMMARIZATION

- **EXTRACTIVE (THE HIGHLIGHTER):** SELECTS EXISTING SENTENCES.
  - PROS: FACTUAL. CONS: DISJOINTED FLOW.
- **ABSTRACTIVE (THE GHOSTWRITER):** GENERATES NEW SENTENCES (SEQ2SEQ).
  - PROS: NATURAL FLOW. CONS: PRONE TO HALLUCINATION.

THE ARCHITECTURE FOR SUMMARIZATION

- **ENCODER-DECODER (T5 / BART / PEGASUS):**
  - ENCODER: READS (UNDERSTANDING).
  - DECODER: WRITES (GENERATION).
- **DECODER-ONLY (GPT-4 / LLAMA 3):**
  - CAN SUMMARIZE VIA "FEW-SHOT PROMPTING", BUT ENCODER-DECODER IS OFTEN MORE EFFICIENT FOR PURE TEXT-TO-TEXT.

NLP & LLM EVALUATION METRICS CHEATSHEET

Comprehensive Guide to Assessing Language Models & Text Generation

TEXT GENERATION QUALITY & FLUENCY

Perplexity

Measures how well a model predicts a sample. Lower perplexity = better performance.

Good For: Language Modeling.
Speed: Fast.

BLEU

Compares n-gram overlap between generated and reference text. Scores 0-1 (Higher is better).

n-gram match

Good For: Machine Translation.
Speed: Very Fast.

ROUGE

(Recall-Oriented Understudy for Gisting Evaluation)

Evaluates summarization via n-gram/sequence overlap. ROUGE-N, ROUGE-L variants.

Recall focus

Good For: Summarization.
Speed: Fast.

METEOR

Improves BLEU with synonyms, stemming, and alignment. Better human correlation.

Good For: Translation, Generation.
Speed: Medium.

CHRF / CHRF++

Character n-gram F-score. Effective for morphologically rich languages.

C-H-A-I-N

Good For: Morphologically Rich Text.
Speed: Fast.

QUESTION ANSWERING & ACCURACY

Exact Match (EM)

Percentage of predictions that exactly match ground truth.

Good For: QA (SQuAD).
Speed: Very Fast.

F1 Score (Token-level)

Precision Recall

Harmonic mean of precision and recall at token level for partial credit.

$$F_1 = 2 \frac{(P * R)}{(P + R)}$$

Good For: QA, NER.
Speed: Fast.

Factuality / Factual Consistency

Assesses if outputs align with factual knowledge. Uses external tools.

Claim

Good For: Verification, Safety.
Speed: Slow (Depends on tools).

SEMANTIC & LEARNED METRICS

BERTScore

cosine similarity

Uses BERT contextual embeddings for similarity. Computes P, R, F1 based on cosine.

$\cos(-\sqrt{2})$ θ

Good For: Semantic Similarity.
Speed: Slow (Requires BERT).

BLEURT

(Bleu Unsupervised Runtime Testing)

Learned evaluation metric trained on human judgments. Robust for fluency and quality.

Training
Human Data → Model → Score

Good For: Fluency, Quality.
Speed: Medium-Slow.

MAUVE

Compares divergence between model and human text distributions. Assesses quality & diversity.

Good Fer: Quality & Diversity.
Speed: Medium.

DIVERSITY, SAFETY & HUMAN EVALUATION

Self-BLEU

Measures diversity by computing BLEU against itself. Lower score = Higher diversity.

Good For: Diversity.
Speed: Fast.

Distinct-n

Unique n-grams divided by total n-grams. Evaluates lexical diversity.

Unique / Total

Good For: Lexical Diversity.
Speed: Fast.

Toxicity Score

Measures harmful or offensive language using classifiers.

Safe Toxic

Good For: Safety.
Speed: Medium (Depends on API).

Human Evaluation

Gold standard. Raters assess fluency, coherence, relevance, and factuality.

1 - 5 - 5

Good For: Validation, Final Check.
Speed: Very Slow & Costly.

Win Rate (Pairwise Comparison)

A/B testing; measures how often one model is preferred over another.

Model A | Model B
Winner: A

Good For: Comparison, Selection.
Speed: Medium-Slow (Requires judging).

SUMMARY: Combine multiple metrics for a holistic assessment. No single metric captures all quality aspects. Balance automatic evaluation with human judgment for reliable results.

EVALUATION: THE ROUGE METRIC

- **WHY ACCURACY DOESN'T WORK:** NO SINGLE "CORRECT" SUMMARY.
- **ROUGE:** MEASURES OVERLAP OF N-GRAMS BETWEEN GENERATED SUMMARY AND HUMAN REFERENCE.
  - **ROUGE-1:** OVERLAP OF SINGLE WORDS.
  - **ROUGE-L:** LONGEST COMMON SUBSEQUENCE (STRUCTURE).

THE HALLUCINATION DANGER

- **EXAMPLE:**
  - SOURCE: "THE ROBOT BATTERY IS AT 10%."
  - BAD SUMMARY: "THE ROBOT IS FULLY CHARGED."
- **MITIGATION:**
  - FACT-CHECKING LAYERS (NATURAL LANGUAGE INFERENCE).
  - LOW TEMPERATURE (REDUCING CREATIVITY).

THE FINE-TUNING PROBLEM

- **CONTEXT:** CUSTOMIZING LLAMA-3-8B TO SPEAK LIKE A PIRATE.
- **FULL FINE-TUNING:**
  - REQUIRES UPDATING ALL 8B WEIGHTS.
  - REQUIRES ~120GB+ VRAM (DO YOU HAVE 4 A100 GPUS?).
- **THE SOLUTION:** PEFT (PARAMETER-EFFICIENT FINE-TUNING).

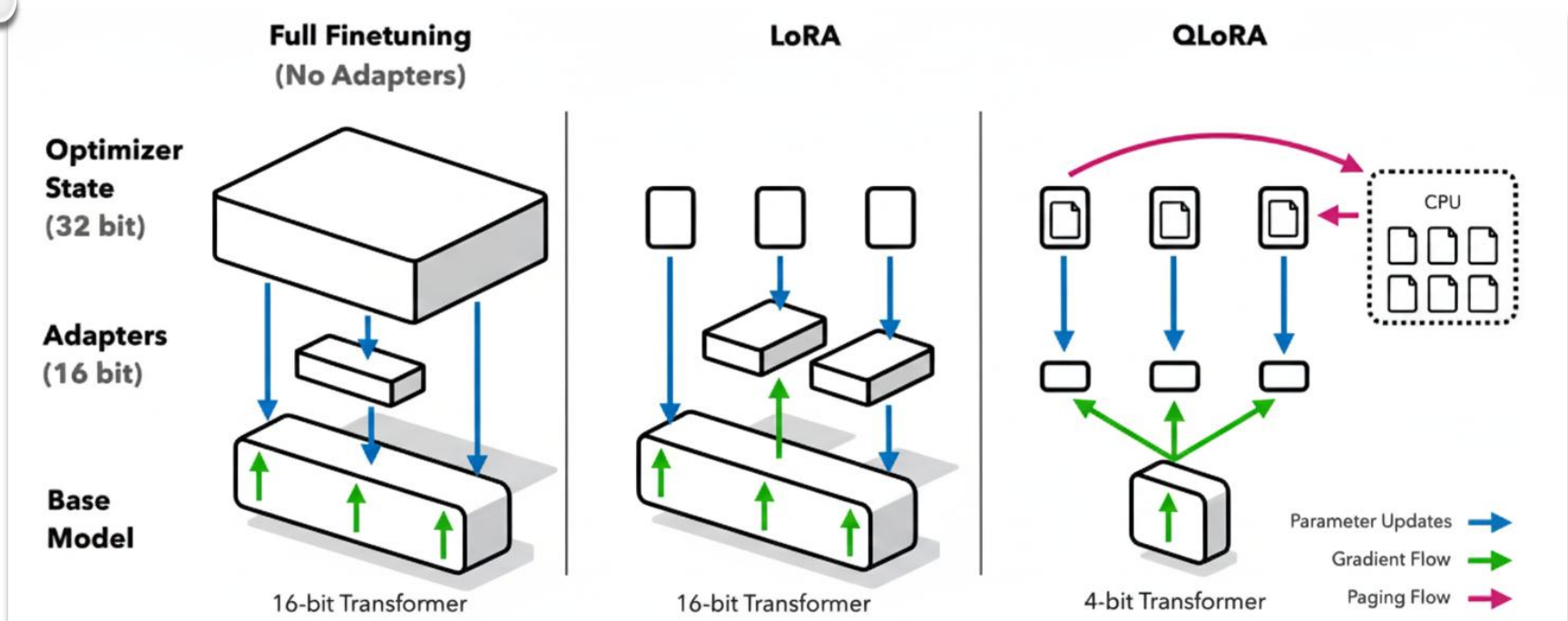
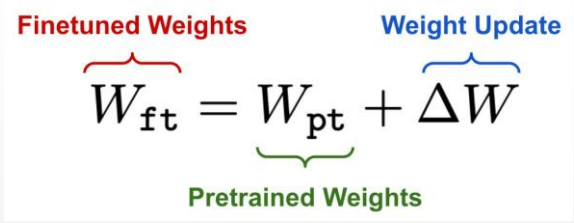


Figure 1: Different finetuning methods and their memory requirements. QLoRA improves over LoRA by quantizing the transformer model to 4-bit precision and using paged optimizers to handle memory spikes.

Source: https://arxiv.org/pdf/2305.14314

ENTER LORA (LOW-RANK ADAPTATION)



- **THE INSIGHT**: WE DON'T NEED TO CHANGE EVERY WEIGHT. THE REQUIRED CHANGE IS LOW-RANK.
- **HOW IT WORKS:**
  - FREEZE THE MASSIVE PRE-TRAINED MODEL (W).
  - INJECT TWO TINY MATRICES (A AND B) NEXT TO LAYERS. TRAIN ONLY A AND B.
- **MATH**: W_NEW = W_FROZEN + (B X A)

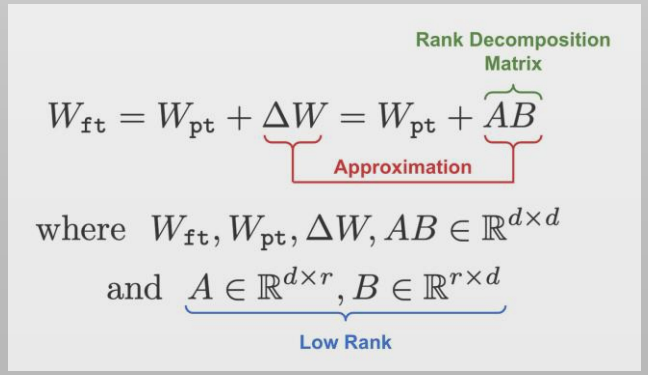
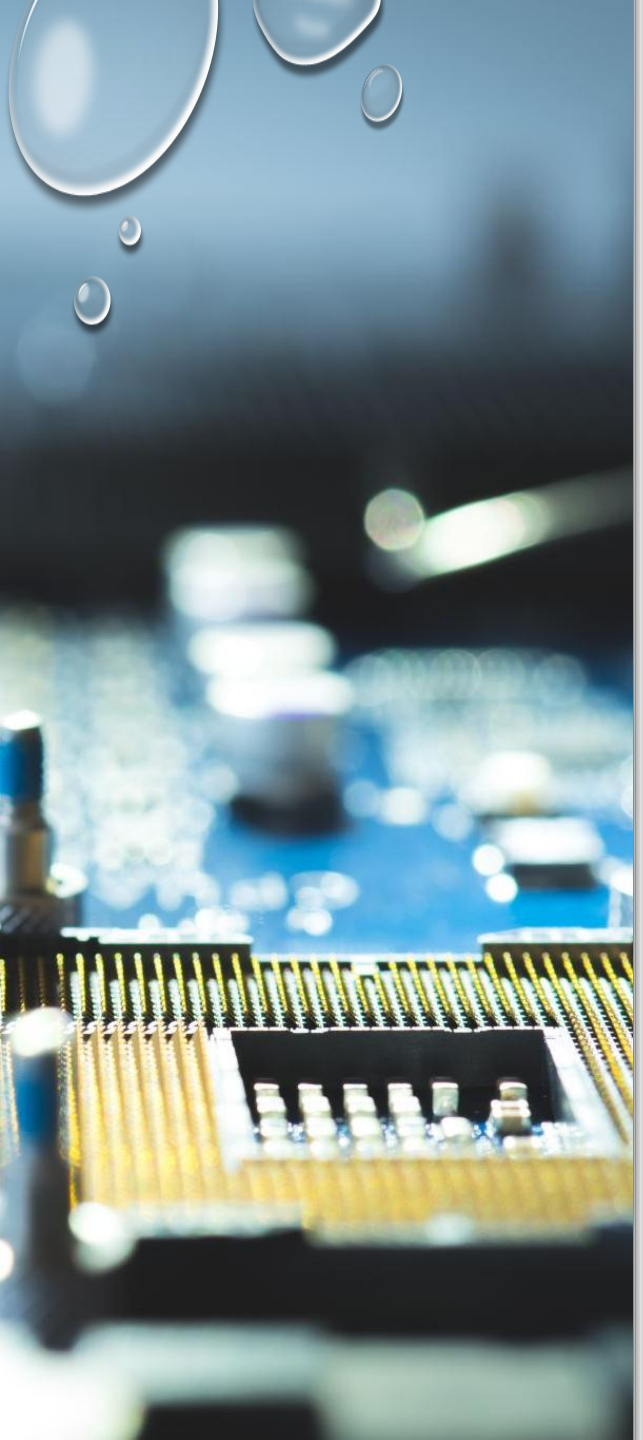


Image Credit: https://cameronrwolfe.substack.com/p/easily-train-a-specialized-llm-peft

THE EFFICIENCY GAINS

- **STORAGE:** FULL LLAMA-3 CHECKPOINT = 15GB. LORA ADAPTER = 20MB.
- **MEMORY:** TRAIN A 7B MODEL ON A SINGLE CONSUMER GPU (RTX 4090 / COLAB).
- **MODULARITY:** SWAP ADAPTERS INSTANTLY.
  - ADAPTER 1: MEDICAL DOCS. ADAPTER 2: LEGAL DOCS.
  - BASE MODEL: STAYS THE SAME.



QLORA (QUANTIZED LORA)

- **2025 STANDARD**:
  - QUANTIZATION: LOAD BASE MODEL IN 4-BIT TO SAVE MEMORY.
  - LORA: APPLY THE ADAPTER ON TOP.
- **RESULT:** FINE-TUNING SOTA MODELS ON CONSUMER HARDWARE.

ADAPTERS IN ROBOTICS

- **THE "SKILL STORE" CONCEPT:**
  - ROBOT BASE BRAIN: GENERAL LANGUAGE UNDERSTANDING.
  - LORA MODULE A: NAVIGATION SYNTAX.
  - LORA MODULE B: TOOL MANIPULATION MANUAL.
- *THE ROBOT LOADS THE SPECIFIC LORA ADAPTER BASED ON THE TASK AT HAND.*

LAB OVERVIEW

- **PART 1: DECODING PLAYGROUND**
  - USE MODEL.GENERATE() WITH DIFFERENT SETTINGS (TEMP 0.1 VS 2.0).
  - OBSERVE HOW BEAM SEARCH FIXES GRAMMAR ERRORS.
  - EVALUATE USING CHEAT SHEET.
- **PART 2: LORA FINE-TUNING**
  - TASK: FINE-TUNE BASE MODEL (FLAN-T5/TINYLLAMA) TO SUMMARIZE DIALOGUES.
  - TOOL: HUGGING FACE PEFT LIBRARY.

SUMMARY

- **CONTROL:** USE NUCLEUS SAMPLING FOR CREATIVITY, BEAM SEARCH FOR ACCURACY.
- **EFFICIENCY:** NEVER FULL FINE-TUNE. USE LORA/PEFT.
- **FUTURE:** ADAPTER FUSION (COMBINING SKILLS).

REFERENCES & READING

- HOLTZMAN ET AL. (2019). "THE CURIOUS CASE OF NEURAL TEXT DEGENERATION" (NUCLEUS SAMPLING). HTTPS://ARXIV.ORG/ABS/1904.09751
- HU ET AL. (2021). "LORA: LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS".
- MIT PAPER ON CHAIN OF DENSITY PROMPTING (2023).
- HUGGING FACE PEFT INDEX

- **NEXT WEEK:** MODULE 7 - RAG SYSTEMS
  RAG SURVEY (2023) - HTTPS://ARXIV.ORG/PDF/2312.10997